A Major Project Report On

COLORECTAL CANCER DETECTION USING PRE-TRAINED ENSEMBLE LEARNING

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering (AI&ML)

Under the Guidance of

Mrs. Shaik Sharmila

Assistant Professor

By

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Department of Computer Science and Engineering (AI&ML)

SRIDEVI WOMEN'S ENGINEERING COLLEGE

(An UGC Autonomous Institution)

(Estd. 2001 | Approved by AICTE & Govt. of TS |Affiliated to JNTUH

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V.N.PALLY, Gandipet, Hyderabad-75

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CERTIFICATE

This is to certify that the **MAJOR PROJECT REPORT** entitled **"Colorectal Cancer Detection Using Pre-Trained Ensemble Learning"** is being submitted by **D.Nikitha(22D21A6608), P.Nandini (22D21A6626), C.Sanjana(22D21A6630)** in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI&ML)is a record of Bonafide work carried out by them.

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Assistant Professor	**Assistant Professor**	**Professor**

EXTERNAL EXAMINER

DECLARATION

We hereby declare that Major Project entitled **"Colorectal Cancer Detection Using Pre-Trained Ensemble Algorithm"** is the work done during the period and the same is submitted in partial fulfillment of the requirements for the award of degree of Bachelor Of Technology in Computer Science and Engineering (AI&ML) from Jawaharlal Nehru Technological University, Hyderabad.

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ABSTRACT

Colorectal cancer (CRC) is one of the leading causes of cancer-related deaths worldwide. Early detection significantly increases the chances of successful treatment and survival. Traditional diagnostic techniques, such as colonoscopy and histopathological examination, although effective, are often invasive, time-consuming, and subject to human error. Recent advancements in machine learning and artificial intelligence, particularly using ensemble algorithms, offer promising tools for automated and accurate detection of colorectal cancer. This project proposes a robust colorectal cancer detection system leveraging pre-trained ensemble algorithms to enhance prediction accuracy and reduce diagnostic latency. By integrating multiple advanced models such as Random Forest, Gradient Boosting, and XGBoost, and fine-tuning them on medical datasets, the system aims to outperform single-model approaches in identifying cancerous patterns from structured or image-based data.

1.INTRODUCTION

1.1.PURPOSE

The main purpose of this project is to create a smart, accurate, and efficient system for the early detection of colorectal cancer using advanced machine learning and artificial intelligence techniques. Colorectal cancer is one of the leading causes of death worldwide, and early detection plays a major role in saving lives. However, traditional methods such as colonoscopy and lab-based tests are often costly, time-consuming, and can sometimes give false or delayed results. To overcome these challenges, this project uses pre-trained ensemble algorithms like Random Forest, Gradient Boosting, and XGBoost, which combine the strengths of multiple models to provide better prediction accuracy and reliability. By training these models on medical datasets, the system can detect hidden patterns that are difficult for humans to identify. This approach helps doctors by providing quick and reliable results, reducing manual workload, minimizing human error, and ensuring that patients receive timely diagnosis and treatment. Ultimately, the project aims to use technology to make cancer detection faster, more precise, and accessible, improving the overall quality of healthcare.

1.2.SCOPE

The scope of this project focuses on developing an advanced and user-friendly system for the early detection of colorectal cancer using pre-trained ensemble machine learning algorithms. The project aims to analyze patient data and medical records to accurately predict the presence of cancer, helping doctors make quick and reliable decisions. By combining models like Random Forest, Gradient Boosting, and XGBoost, the system improves accuracy and reduces errors that occur with traditional single-model or manual methods. This project can be used by healthcare professionals, hospitals, and diagnostic centers as a supportive tool for early diagnosis. In the future, the system can be expanded to include different types of data such as histopathology images, genetic information, and clinical history to further improve its performance. It can also be integrated into hospital management systems or web-based applications for real-time cancer screening and result reporting. Overall, the project provides a strong base for building smarter, faster, and more efficient medical diagnostic systems.

1.3.DEFINITIONS,ABBREVATIONS AND MODEL DIAGRAM

1.3.1 Definitions

1. Colorectal Cancer (CRC): A type of cancer that starts in the colon or rectum. It usually begins as small growths called polyps that can turn into cancer if not removed early.

2. Machine Learning (ML): A branch of artificial intelligence that allows computers to learn from data and make predictions or decisions without being explicitly programmed.

3. Artificial Intelligence (AI): A field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence, such as reasoning, learning, and problem-solving.

4. Ensemble Learning: A machine learning technique that combines multiple models to make more accurate and reliable predictions than a single model.

5. Pre-Trained Model: A model that has already been trained on a large dataset and can be reused or fine-tuned for a new task, saving time and resources.

6. Data Flow Diagram (DFD): A visual representation that shows how data moves through a system, including inputs, processes, and outputs.

7. Relational Database Management System (RDBMS): A type of database that organizes data into tables with rows and columns, allowing easy access and management using SQL.

1.3.2 Abbreviations

- **CRC** – Colorectal Cancer
- **XAI** – Explainable Artificial Intelligence
- **RF** – Random Forest
- **GB** – Gradient Boosting
- **XGB** – XGBoost (Extreme Gradient Boosting)

1.3.3 .Model Diagram

A Data Flow Diagram (DFD), also called a bubble chart, is a simple graphical tool used to represent a system. It shows the input data, the processing performed on this data, and the output generated. DFDs are important for modeling system components, which include the processes, the data used, external entities interacting with the system, and the information flows. They depict how information moves through the system and how it is transformed from input to output. DFDs can be drawn at any level of abstraction and can be divided into multiple levels to show increasing detail about the system's functions and data flow.

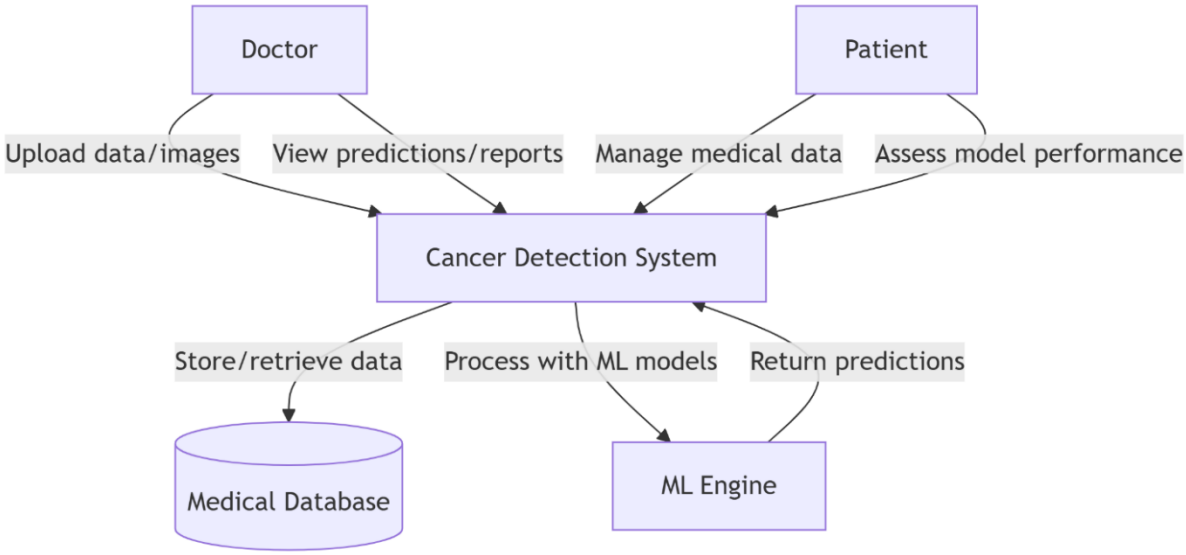


Figure: 1.1 Model Diagram

1.3.4 OVERVIEW

This project develops an automated system to detect colorectal cancer using pre-trained ensemble models like Random Forest, Gradient Boosting, and XGBoost. It analyzes both clinical data histopathology images to provide accurate, fast, and reliable predictions, along with visual explanations and diagnostic reports. The system aims to assist doctors in early detection, reduce errors, and improve patient care, while being scalable and user-friendly for future enhancement.

2. LITERATURE SURVEY

Title: "Machine Learning Applications in Colorectal Cancer Diagnosis: A Review"

Abstract: Machine learning has transformed cancer diagnostics through automated classification and prediction. Random Forest and Support Vector Machines demonstrate promising results in analyzing histopathological images and clinical data for colorectal cancer detection. Single-model approaches face limitations including overfitting and poor generalizability. Ensemble learning strategies that combine multiple models offer more robust and reliable performance for clinical applications.

Title: "Deep Learning Approaches for Colorectal Cancer Detection in Histopathological Images"

Abstract: Convolutional neural networks have proven effective in analyzing histopathology slides for colorectal cancer detection. CNNs outperform traditional handcrafted feature methods by automatically learning discriminative patterns from images. Despite their effectiveness, deep learning models demand large datasets and substantial computational resources. Hybrid approaches integrating CNNs with ensemble algorithms address these challenges and achieve superior accuracy in cancer classification.

Title: "Ensemble Learning Methods in Colorectal Cancer Diagnostics"

Abstract: Ensemble learning combines multiple classifiers such as Random Forest, Gradient Boosting, and XGBoost to enhance diagnostic accuracy in medical applications. For colorectal cancer detection, ensemble models surpass traditional single classifiers by utilizing complementary strengths and minimizing variance. Results indicate that ensemble learning provides a reliable, robust, and scalable framework for early cancer detection suitable for clinical deployment

2.1 TECHNOLOGIES USED

TECHNOLOGY	CATEGORY	ROLE
Python 3.10	Programming language	Core coding language for implementation.
Django	Backend Framework	Connects models with web interface.
Random forest, Gradient Boosting, XGBoost, AdaBoost	Ensemble Algorithms	Main models for cancer detection.
MySQL	Database	Stores patient data and result.
HTML5, CSS3, JavaScript	Front-end	Builds interactive user interface.
Transfer Learning	ML Technique	Improves accuracy and reduces training time.
Feature Importance Analysis	Explainable AI	Shows key features influencing prediction

3.SYSTEM ANALYSIS

3.1.EXISTING SYSTEM AND DISADVANTAGES

Existing System

In the existing colorectal cancer detection systems, conventional machine learning techniques such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees are predominantly used. These models often rely on manually selected features extracted from patient records or medical imaging, which may limit their ability to generalize across diverse populations or datasets. Moreover, the performance of single models tends to vary significantly with changes in data quality or sample size. In clinical settings, these systems often suffer from poor interpretability, scalability issues, and high false positive or false negative rates, making them unreliable for critical diagnostic tasks. Additionally, these traditional approaches require extensive preprocessing and domain expertise for feature engineering, increasing development time and complexity. They also struggle to handle large-scale medical datasets efficiently, leading to slower prediction times in real-world environments. As a result, their practical deployment in hospitals and large diagnostic centers becomes challenging.

Disadvantages of Existing System

1.Limited Accuracy: Single models may miss complex patterns in medical data, which can lower prediction accuracy. This makes them less reliable for detecting subtle signs of colorectal cancer.

2.Manual Feature Engineering: The system relies on experts to select important features, which is time-consuming and may introduce errors. This also limits the model's adaptability to new data.

3.Poor Generalization: These models may not perform well on different patients or datasets, reducing their usefulness in real-world settings.

4.High Error Rates: They often produce false positives or false negatives, which can lead to misdiagnosis or unnecessary tests.

5.Hard to Understand: Clinicians may find it difficult to interpret how the system makes decisions, affecting trust in the results.

3.2.PROBLEM STATEMENT

Colorectal cancer is dangerous, and early detection can save lives. Traditional tests are slow, uncomfortable, and can make mistakes. Current AI methods often use a single model, which is not always accurate. This project aims to build a smart system using multiple pre-trained AI models to detect cancer quickly, accurately, and clearly, helping doctors make better decisions.

3.3.PROPOSED SYSTEM AND ADVANTAGES

The proposed system utilizes pre-trained ensemble learning algorithms to detect colorectal cancer with enhanced accuracy, efficiency, and reliability. Ensemble methods combine the predictions of multiple machine learning models to create a more robust and generalized system compared to single-model approaches. Techniques such as Random Forest, AdaBoost, and Gradient Boosting are fine-tuned using transfer learning and trained on comprehensive datasets that include structured clinical data and histopathological images, enabling the system to learn complex and non-linear patterns associated with colorectal cancer. This integrated approach improves classification performance by capturing subtle relationships within medical data that traditional models may fail to detect, while also reducing overfitting and increasing generalization across diverse patient datasets. The use of pre-trained models minimizes training time and computational cost by leveraging knowledge gained from previously trained large-scale datasets. Additionally, cross-validation and hyperparameter optimization techniques are applied to ensure optimal model performance and stability. Feature importance analysis and explainable AI methods are incorporated to enhance interpretability, allowing healthcare professionals to understand the key factors influencing predictions and increasing trust in the system. Overall, the proposed system serves as an intelligent, scalable, and efficient diagnostic support tool that assists doctors in early detection, reduces human error, and contributes to improved patient care and survival outcomes.

Advantages of Proposed System

1. High Accuracy- The system combines multiple pre-trained AI models to detect colorectal cancer more reliably than single-model approaches. It captures subtle patterns in clinical and image data, reducing misclassification.

2. Fast Detection- Using pre-trained models reduces training time and allows quicker predictions. This helps clinicians get timely results for early diagnosis.

3. Handles Diverse Data- It can process both structured medical records and microscopic images. This makes the system versatile and capable of comprehensive analysis.

4. Interpretability- Feature importance and visual explanations like heatmaps help doctors understand the basis of predictions. This builds trust in the system's recommendations.

5.Improved Generalization – The ensemble approach reduces overfitting and performs consistently across different datasets and patient groups. This ensures stable and reliable predictions in real-world clinical environments.

6.Scalability and Efficiency – The system is designed to handle large volumes of medical data efficiently, making it suitable for deployment in hospitals and large diagnostic centers without performance degradation.

4.SYSTEM REQUIREMENT SPECIFICATIONS

4.1. FUNCTIONAL REQUIREMENTS

Data Input and Management : The system allows users to enter patient medical data such as lab results, clinical observations, and medical history.

Results and Reporting : Predictions are shown with clear visual indicators and confidence scores for easy understanding.

Admin Functions : Administrators can upload and organize medical datasets within the system. They can retrain AI models whenever new data is available.

User Management : The system uses role-based access control to ensure secure and restricted access. Only authorized users, such as doctors and administrators, can use certain features.

4.2.NON-FUNCTIONAL REQUIREMENTS

Performance: The system should give predictions quickly, so doctors don't have to wait long for results.

Scalability: It should be able to handle more patient data and images as the system grows without slowing down.

Reliability: The system must work consistently and be available whenever needed.

Security: Patient data should be safe and protected from unauthorized access.

Usability: The system should be easy to use, with clear instructions and simple navigation for doctors and admins.

4.3.SOFTWARE REQUIREMENTS

Operating System	Windows 10/11
Development Software	Python 3.10
Programming language	Python
Front End Technologies	HTML,CSS3, Java Script
Back End Technologies	Django
Database Language	SQL
Database(RDBMS)	MySQL
Web Server or Deployment Server	Django Application Development Server

4.4.HARDWARE REQUIREMENTS

MINIMUM (Required for Execution)	MY SYSTEM (Development)	
System	Pentium IV 2.2 GHz	i3 Processor 5th Gen
Hard Disk	20 Gb	500 Gb
Ram	1 Gb	4 Gb

5.SYSTEM DESIGN

5.1.SYSTEM SPECIFICATIONS

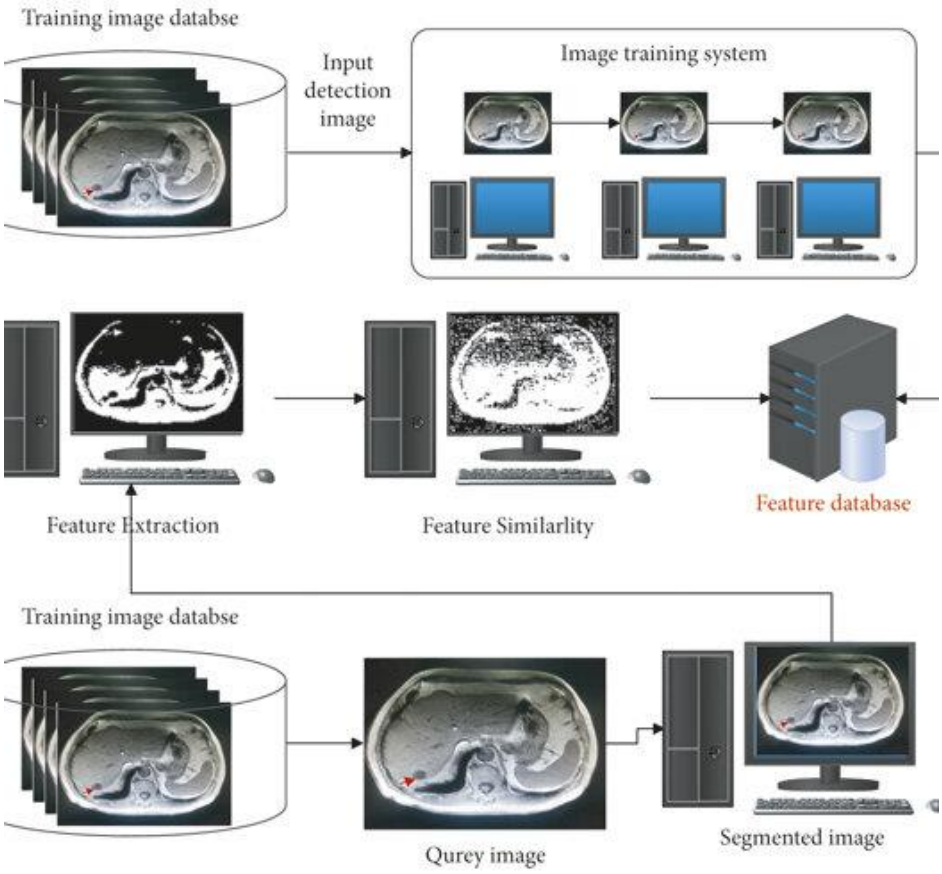


Figure: 5.1 System Architecture

- The system follows a layered approach with a clear separation between the front-end (HTML, CSS, JavaScript), application logic (Python-based processing using Django), and back-end (MySQL database).
- The backend handles data management, model training, prediction, and report generation, processing both structured clinical data and histopathology images.
- Each function—such as data preprocessing, feature extraction, ensemble model integration, prediction, and report generation—is managed by a dedicated module for better scalability and maintainability.
- The system supports secure storage and access to sensitive patient data, ensuring authentication, authorization, and encryption, while maintaining fast and reliable performance.

5.2.SYSTEM COMPONENTS

Admin modules

Data Acquisition and Preprocessing: Admins can upload structured clinical data and histopathology images. The system handles data cleaning, normalization, encoding, and augmentation automatically.

Feature Extraction and Selection: The system extracts relevant features from both clinical and image data. Feature selection techniques are applied to reduce unnecessary data and improve model performance.

Ensemble Learning Model Integration: Multiple machine learning models, like Random Forest, Gradient Boosting, and XGBoost, are integrated. Pre-trained models are fine-tuned, and ensemble strategies like stacking or majority voting are applied to improve accuracy.

Model Training and Optimization: The system trains models using cross-validation, performs hyperparameter tuning, and monitors sensitivity and specificity to optimize performance.

Evaluation and Validation: Admins can evaluate model performance using metrics like accuracy, precision, recall, F1-score, and AUC-ROC. The system can analyze confusion matrices and validate models on external datasets.

User Modules

User Interface and Data Input Module: Doctors can input patient information and upload histopathology images. The system provides real-time feedback on data quality, such as missing or incorrect entries.

Prediction and Diagnostic Assistance Module: The trained ensemble model generates predictions for the input data. Confidence scores and visual explanations, such as heatmaps for images, help clinicians understand the results.

Report Generation and Export Module: The system can generate downloadable reports summarizing predictions, confidence levels, and important features. These reports assist doctors in making informed decisions.

5.3.UML DIAGRAMS

1.Usecase diagram

The use case diagram illustrates how Admins manage clinical data and histopathology images, perform feature extraction, train and validate ensemble models, and control user access. Doctors input patient data, receive predictions with confidence scores and visual explanations, and generate diagnostic reports. The system connects all modules seamlessly, providing accurate, fast, and interpretable colorectal cancer detection to support clinical decision-making.

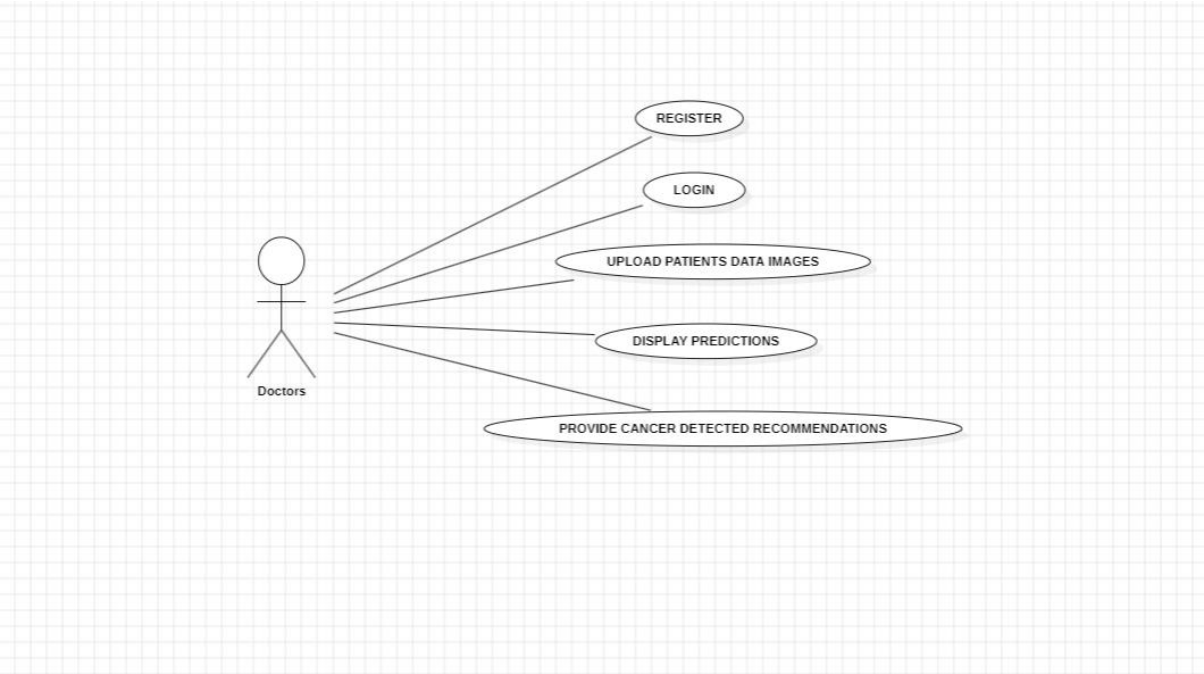


Figure: 5.2 Usecase Diagram(Doctor)

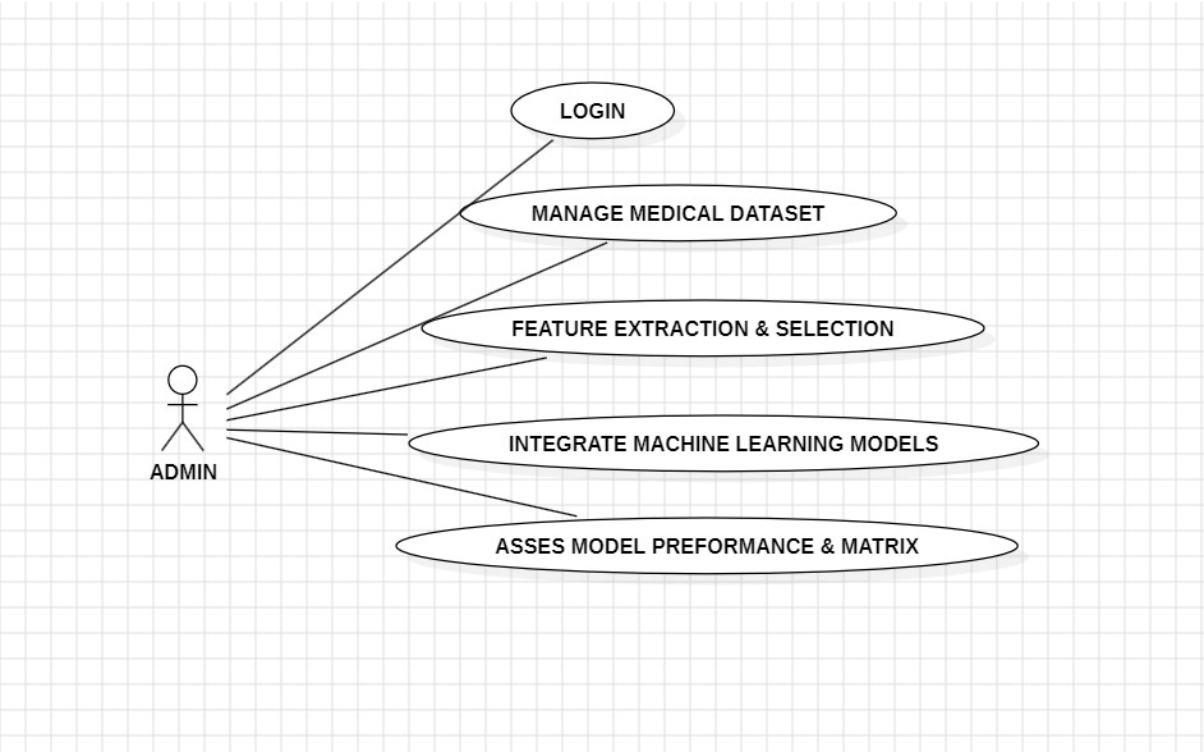


Figure: 5.3 Usecase Diagram (Admin)

2. Class diagram

The class diagram represents the static structure of the colorectal cancer detection system by showing the system's main classes, their attributes, methods, and relationships. Key classes include Admin, Doctor/User, Patient Data, Ensemble Model, and Report. Admin handles data management, model training, and validation, while Doctor/User interacts with Patient Data and requests predictions. Ensemble Model performs preprocessing, feature extraction, model integration, and prediction generation, and the Report class manages the creation and export of diagnostic reports. The diagram highlights how these classes interact and depend on each other to ensure seamless functionality.

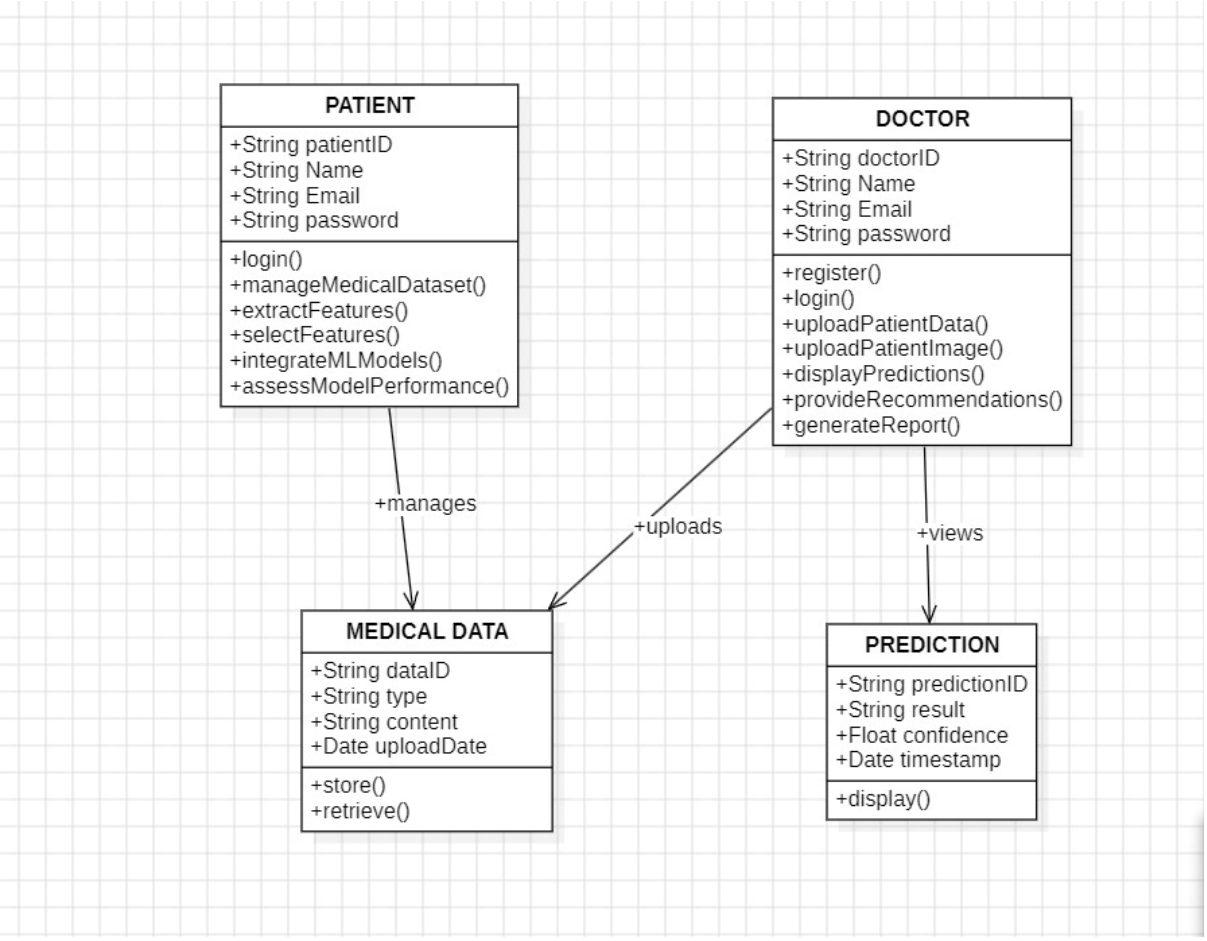


Figure: 5.4 Class Diagram

3.Sequence Diagram

The sequence diagram illustrates the dynamic flow of interactions between users and the system over time. When a Doctor inputs patient data, the system first preprocesses the data, performs feature extraction, runs the trained ensemble models for prediction, and returns confidence scores and visual explanations. The Doctor can then request a report, which is generated and delivered. Admin interactions follow a similar sequence, including data upload, model training, validation, and monitoring, showing step-by-step message exchanges between users, system modules, and the database.

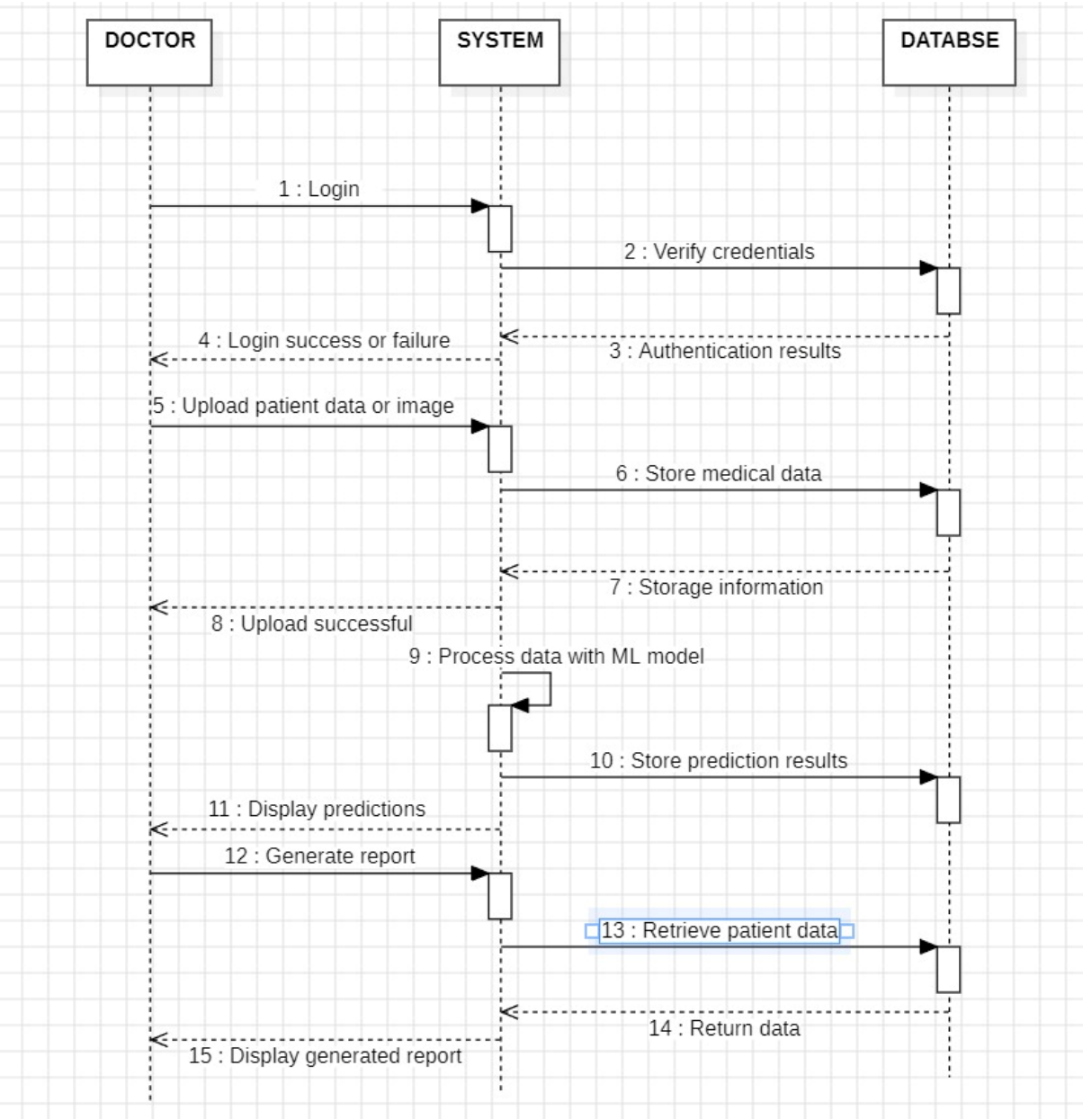


Figure: 5.5 Sequence Diagram

4.Activity Diagram

The activity diagram represents the workflow of the colorectal cancer detection process, capturing all major actions and decision points. It starts with data acquisition, followed by preprocessing and feature extraction. The system then trains or applies ensemble models to make predictions. Based on the prediction outcome, the system generates visual explanations and reports. Admin activities like model training, validation, and user management, as well as Doctor activities like data input and report generation, are represented to show the flow of control and interaction across the system.

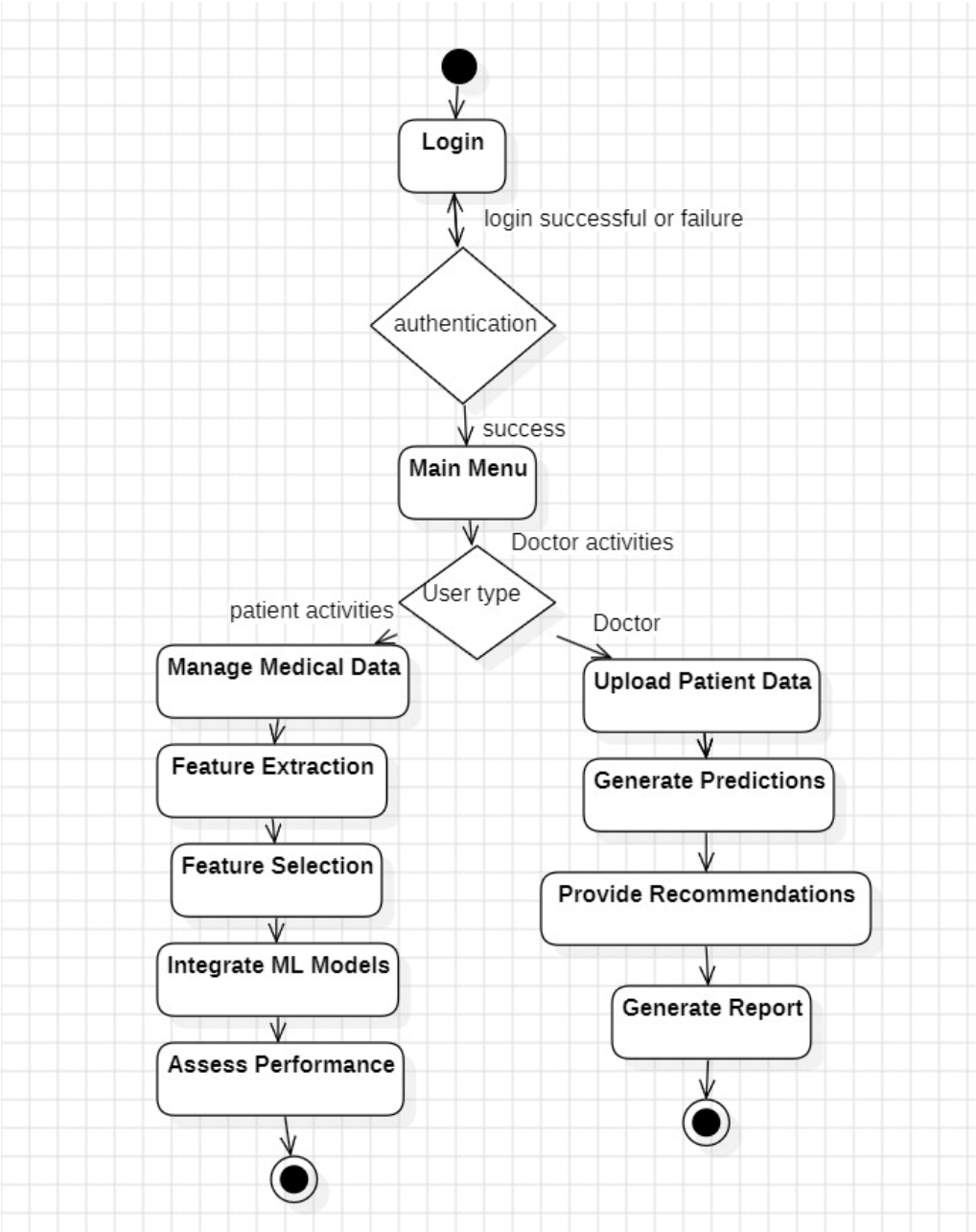


Figure: 5.6 Activity Diagram

6.IMPLEMENTATION

6.1.SAMPLE CODE

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required, user_passes_test
from django.http import JsonResponse
from django.contrib import messages
from django.core.files.base import ContentFile
from .models import PatientData, Prediction, EnsembleModel, ModelTrainingJob, Report
from .forms import PatientDataForm, ModelTrainingJobForm
from .ml_models.ensemble_model import EnsembleModelManager
import cv2, numpy as np
def admin_required(fn):
 return user_passes_test(lambda u: u.is_active and u.is_staff, login_url='/login/')(fn)
@login_required
def patient_input(request):
form = PatientDataForm(request.POST or None, request.FILES or None)
if request.method == "POST" and form.is_valid():
data = form.save(commit=False)
data.uploaded_by = request.user
data.save()
model = EnsembleModel.objects.filter(is_active=True).first()
if not model:
messages.error(request, "No active model available.")
return redirect("patient_input")
prediction = make_prediction(data, model)
return redirect("prediction_result", prediction_id=prediction.id)
return render(request, "user/patient_input.html", {"form": form})
def make_prediction(patient_data, model):
res = EnsembleModelManager().predict(patient_data)
return Prediction.objects.create(
patient_data=patient_data,
ensemble_model=model,
```

```
predicted_class=res["class"],
confidence_score=res["confidence"],
prediction_details=res["details"],
)
@login_required
def prediction_result(request, prediction_id):
pred = get_object_or_404(Prediction, id=prediction_id)
if pred.patient_data.uploaded_by != request.user and not request.user.is_staff:
messages.error(request, "Access denied.")
return redirect("home")
heatmap = generate_heatmap_data(pred.patient_data.histopathology_image.path) \
if pred.patient_data.histopathology_image else None
return render(request, "user/prediction_result.html",
{"prediction": pred, "heatmap_data": heatmap})
def generate_heatmap_data(path):
img = cv2.imread(path)
return np.random.rand(img.shape[0], img.shape[1]).tolist()
@login_required
@admin_required
def training_jobs(request):
form = ModelTrainingJobForm(request.POST or None)
if request.method == "POST" and form.is_valid():
job = form.save(commit=False)
job.status = "pending"
job.save()
from .tasks import train_model_task
train_model_task.delay(job.id)
messages.success(request, "Training started!")
return redirect("training_jobs")
jobs = ModelTrainingJob.objects.all().order_by("-created_at")
return render(request, "admin/training_jobs.html", {"form": form, "jobs": jobs})
def api_training_progress(request, job_id):
job = get_object_or_404(ModelTrainingJob, id=job_id)
```

```
return JsonResponse({"status": job.status, "accuracy": job.accuracy, "log": job.training_log})
@login_required
def generate_report(request, prediction_id):
pred = get_object_or_404(Prediction, id=prediction_id)
report = Report.objects.create(
prediction=pred,
report_file=generate_pdf_report(pred)
)
return redirect("download_report", report_id=report.id)
def generate_pdf_report(pred):
text = f"""
Colorectal Cancer Diagnosis Report
Patient: {pred.patient_data.patient}
Date: {pred.created_at}
Prediction: {pred.get_predicted_class_display()}
Confidence: {pred.confidence_score:.2%}
Model Used: {pred.ensemble_model.name}
"""
return ContentFile(text.encode("utf-8"), name=f"report_{pred.id}.txt")
```

7.SYSTEM TESTING

7.1.TESTING STRATEGIES

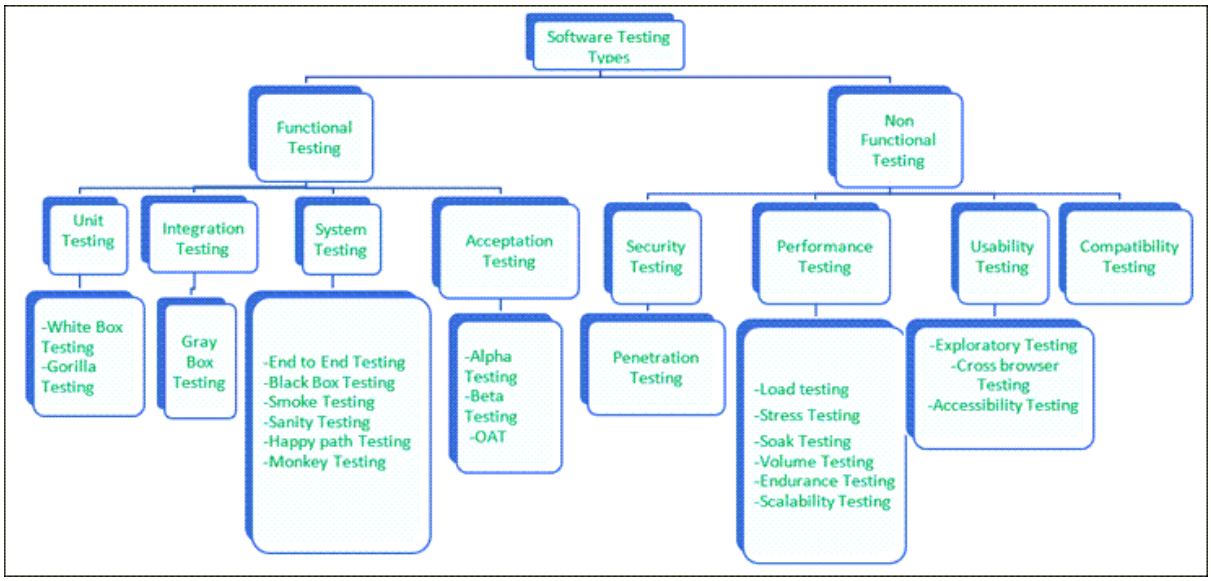


Figure: 7.1 Testing Strategies

Functional Testing

1.Unit testing

Unit testing is performed on individual modules of the colorectal cancer detection system to verify their correctness. Each module such as data preprocessing, feature extraction, ensemble model prediction, heatmap generation, and report generation is tested separately. Developers check whether each function returns accurate outputs for given inputs. Unit testing helps in identifying defects at an early stage and ensures that every component works independently without errors.

2.Integration Testing

Integration testing is carried out by combining two or more modules and testing them together. In this project, modules such as data input, preprocessing pipeline, ensemble model execution, database storage, and result visualization are integrated and tested. The main objective is to verify proper communication and data flow between modules. It ensures that patient data flows correctly from input to prediction and report generation without loss or corruption.

3.System Testing

System testing evaluates the complete colorectal cancer detection system against specified requirements. The entire workflow from user login, image upload or clinical data entry, model prediction, heatmap visualization, and report download is tested as a whole.

a.End-to-End Testing: This involves testing the entire application in a real-world scenario, including database interaction, model execution, and UI display.

b.Black Box Testing: Testing is performed based only on inputs and expected outputs without considering internal code structure.

4.Acceptance testing

Acceptance testing is conducted to ensure that the system meets clinical requirements and user expectations. Doctors or authorized users test the application using real or sample patient data. The system is validated based on accuracy of cancer prediction, clarity of confidence scores, and usefulness of generated reports.

a.Alpha Testing: Performed internally to detect defects before deployment.

b.Beta Testing: Performed in a real environment with limited users to confirm system reliability before full-scale implementation.

Non-Functional Testing

1.Security Testing

Security testing ensures that sensitive patient data is protected from unauthorized access. Authentication, role-based access control, and secure database handling mechanisms are tested to prevent data breaches.

2.Performance Testing

Performance testing evaluates the system's stability and response time under different workloads. It checks how quickly the ensemble model processes medical data and generates predictions. The system should provide results within acceptable time limits even when handling large datasets.

3.Usability Testing

Usability testing verifies that the user interface is simple, clear, and easy to operate for doctors and administrators. It ensures smooth navigation, proper visualization of results, and easy report download functionality.

7.2. TEST CASES

Table 7.1 Test case for Data Preprocessing Validation Test

Test Cases Status	Priority(H,L): High	
Test Objective	Verify data preprocessing works correctly	
Test Description	Ensure uploaded data is cleaned and processed without errors	
Requirements Verified	Data cleaning, missing value handling	
Test Environment	Web Application – Admin Module	
Test Setup	Admin uploads dataset into the system	
Inputs	Expected Result	Actual Result
Upload raw dataset	Data processed without errors	Data processed successfully
Dataset with missing values	Missing values handled correctly	Missing handled properly
Pass: **Yes**	Conditional Pass: No	Fail: No

Table 7.2 Test Case for Ensemble Model Integration

Test Cases Status	Priority(H,L): High	
Test Objective	Verify ensemble model integration	
Test Description	Ensure multiple models are combined correctly	
Requirements Verified	Model integration and prediction improvement	
Test Environment	Web Application – Admin Module	
Test Setup	Admin integrates trained models	
Inputs	Expected Result	Actual Result
Combine multiple models	Ensemble model created successfully	Ensemble created
Run prediction	Improved performance	Performance improved
Pass: **Yes**	Conditional Pass: No	Fail: No

Table 7.3 Test case for Model Evaluation Metrics

Test Cases Status	Priority(H,L): High	
Test Objective	Verify evaluation metrics generation	
Test Description	Ensure accuracy, precision, recall, and F1-score are calculated	
Requirements Verified	Metric calculation and validation	
Test Environment	Web Application – Admin Module	
Test Setup	Admin runs evaluation process	
Inputs	Expected Result	Actual Result
Validation dataset	Metrics generated correctly	Metrics displayed
External dataset	Acceptable performance level	Performance within limit
Pass: **Yes**	Conditional Pass: No	Fail: No

Table 7.4 Test case for User Input Validation

Test Cases Status	Priority(H,L): High	
Test Objective	Verify user input validation	
Test Description	Ensure incorrect inputs are detected and corrected	
Requirements Verified	Field validation and error messages	
Test Environment	Web Application – User Module	
Test Setup	User enters data for prediction	
Inputs	Expected Result	Actual Result
Missing fields	Error message shown	Error displayed
Correct inputs	Data accepted successfully	Data submitted
Pass: **Yes**	Conditional Pass: No	Fail: No

Table 7.5 Test case for Prediction and Report Generation Test

Test Cases Status	Priority(H,L): High	
Test Objective	Verify prediction and report generation	
Test Description	Ensure prediction result and report are generated correctly	
Requirements Verified	Prediction output, confidence score, report download	
Test Environment	Web Application – User Module	
Test Setup	User submits data and requests report	
Inputs	Expected Result	Actual Result
Valid input data	Prediction displayed with confidence	Prediction shown
Generate report	Report downloaded successfully	Report downloaded
Pass: **Yes**	Conditional Pass: No	Fail: No

7.3 RESULTS AND DISCUSSIONS

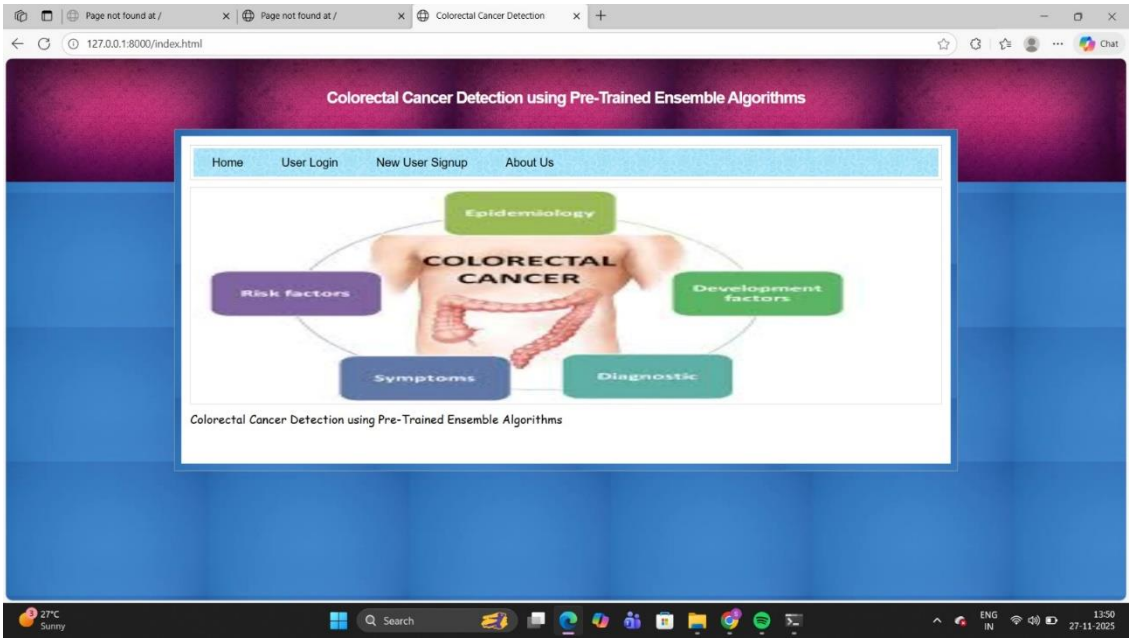


Figure 7.2 Home Screen

This is the main homepage screen with navigation options such as Home, User Login, Signup, and About Us. It also displays the colorectal cancer illustration under the header.

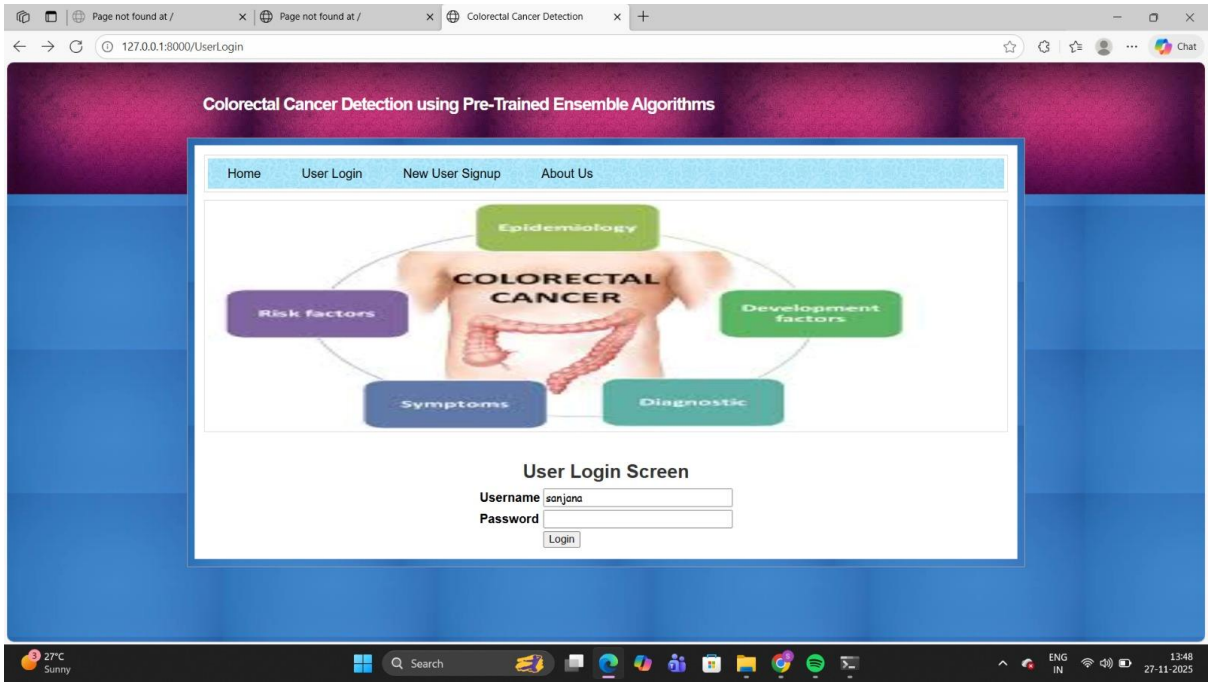


Figure 7.3 Login Page

The user login screen allows entry of username and password for authentication. The colorectal cancer illustration appears above the login form.

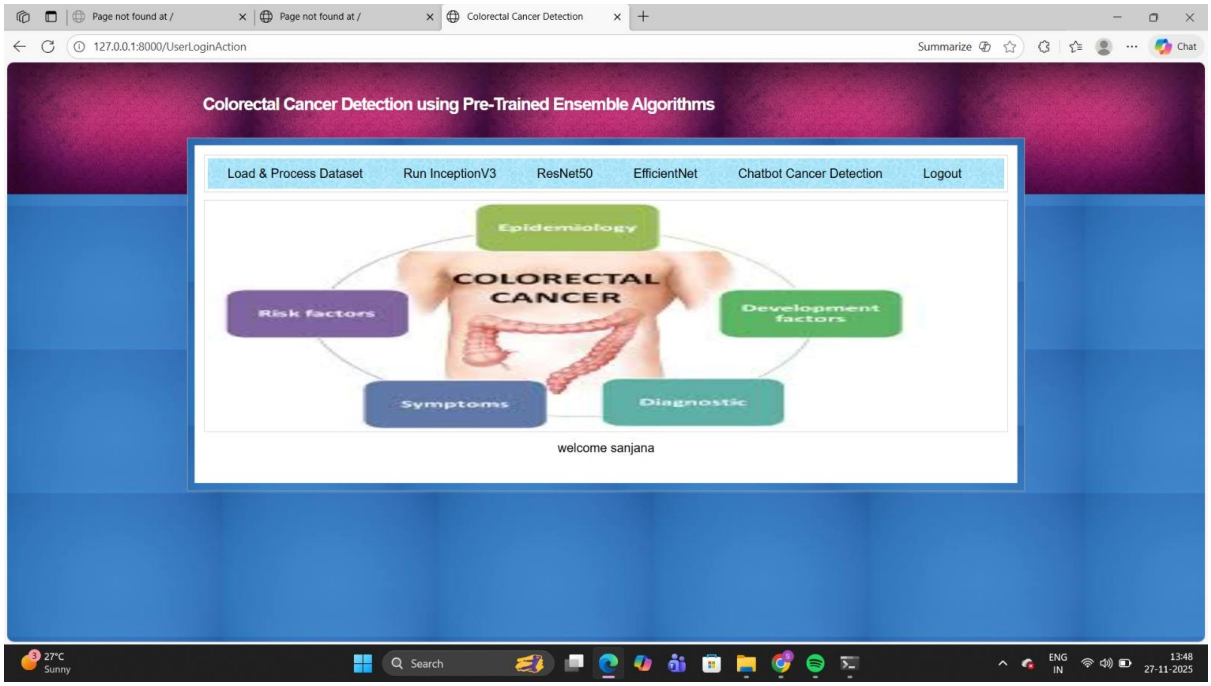


Figure 7.4 Login successful page

A colorectal cancer detection dashboard homepage showing main navigation options like dataset processing and model execution. The screen welcomes the logged-in user “sanjana” with an illustrative cancer diagram.

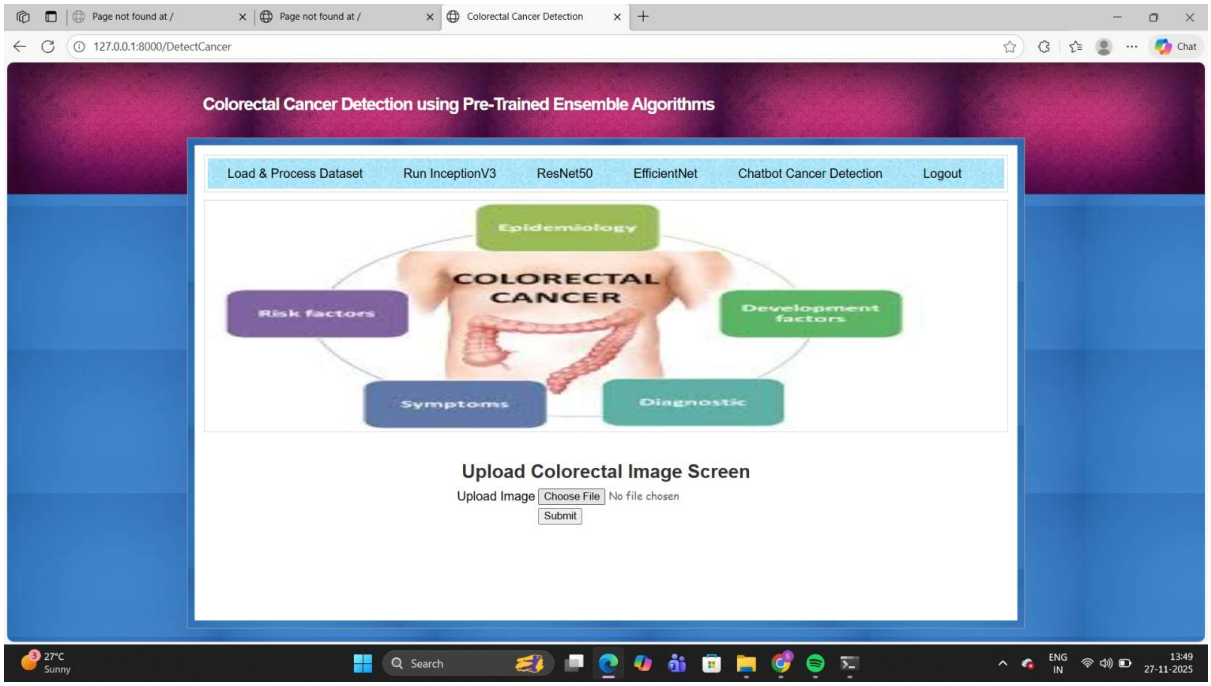


Figure 7.5 Upload scan image page

The "Upload Colorectal Image Screen" page shows the file upload field and submit button. The background contains the main colorectal cancer illustration and menu bar.

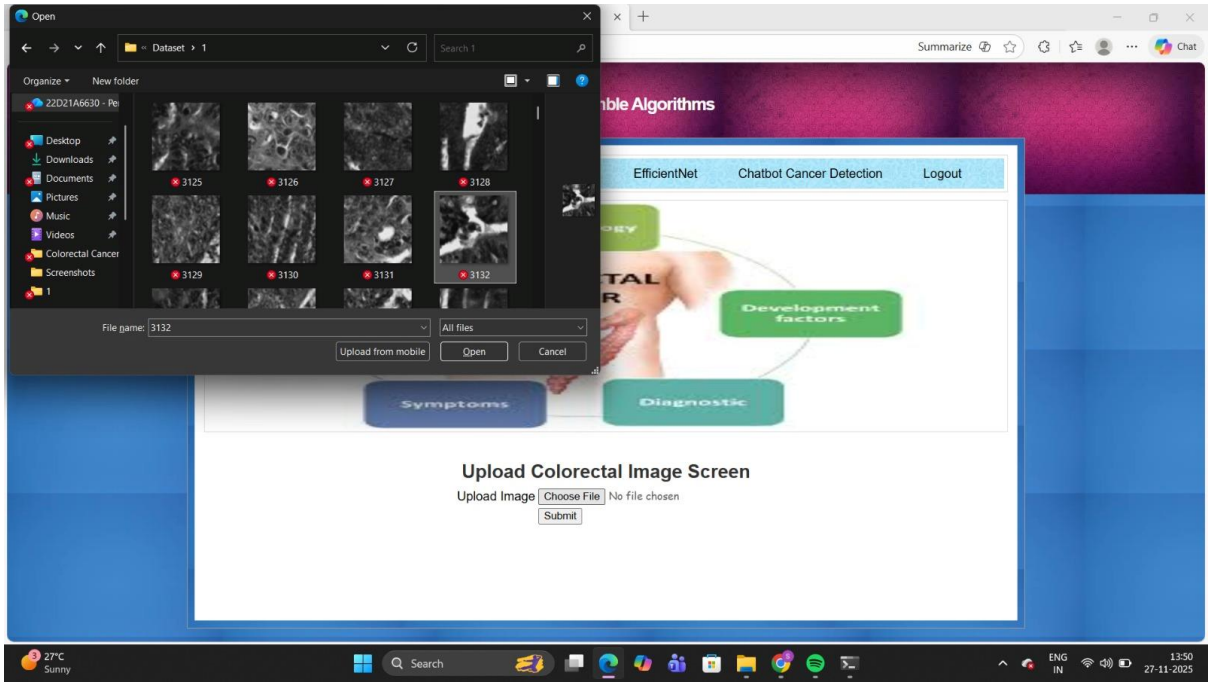


Figure 7.6 Image Retrieval page

The image upload dialog is open, allowing the user to select a colorectal image from the dataset folder. Various grayscale sample images are shown for selection.

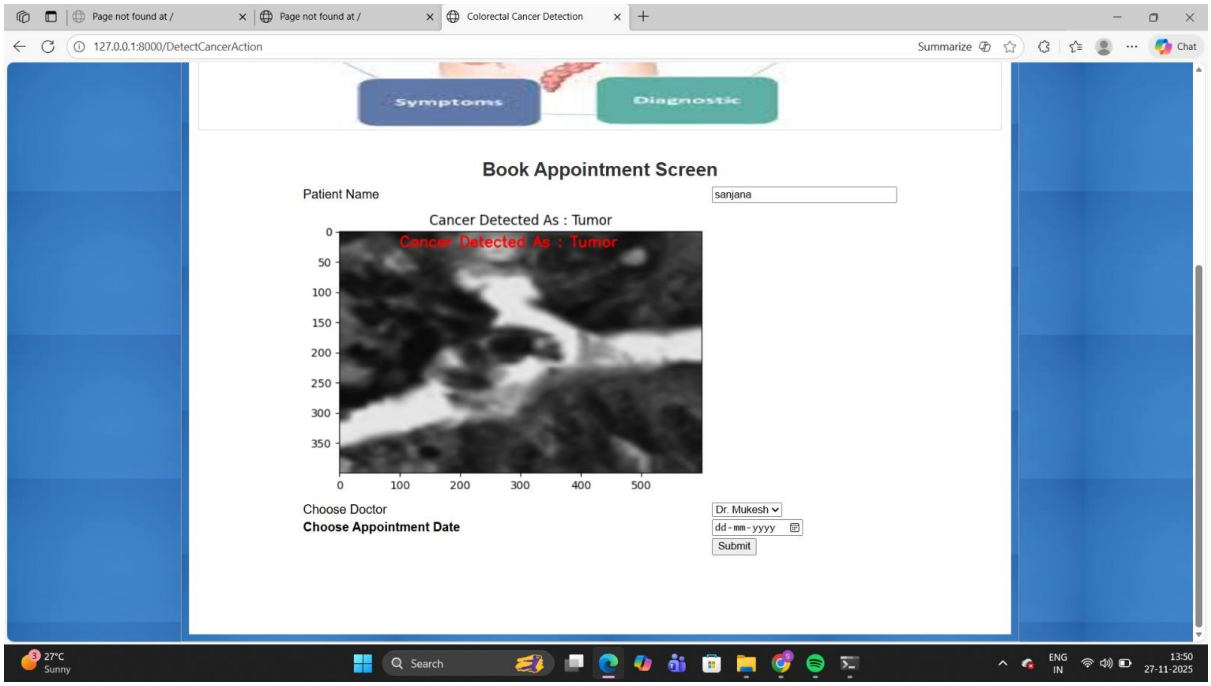


Figure 7.7 Cancer Detection Image page

The appointment booking screen displays the uploaded colorectal image with the prediction "Cancer Detected As: Tumor." The user can select a doctor, choose a date, and submit the appointment form.

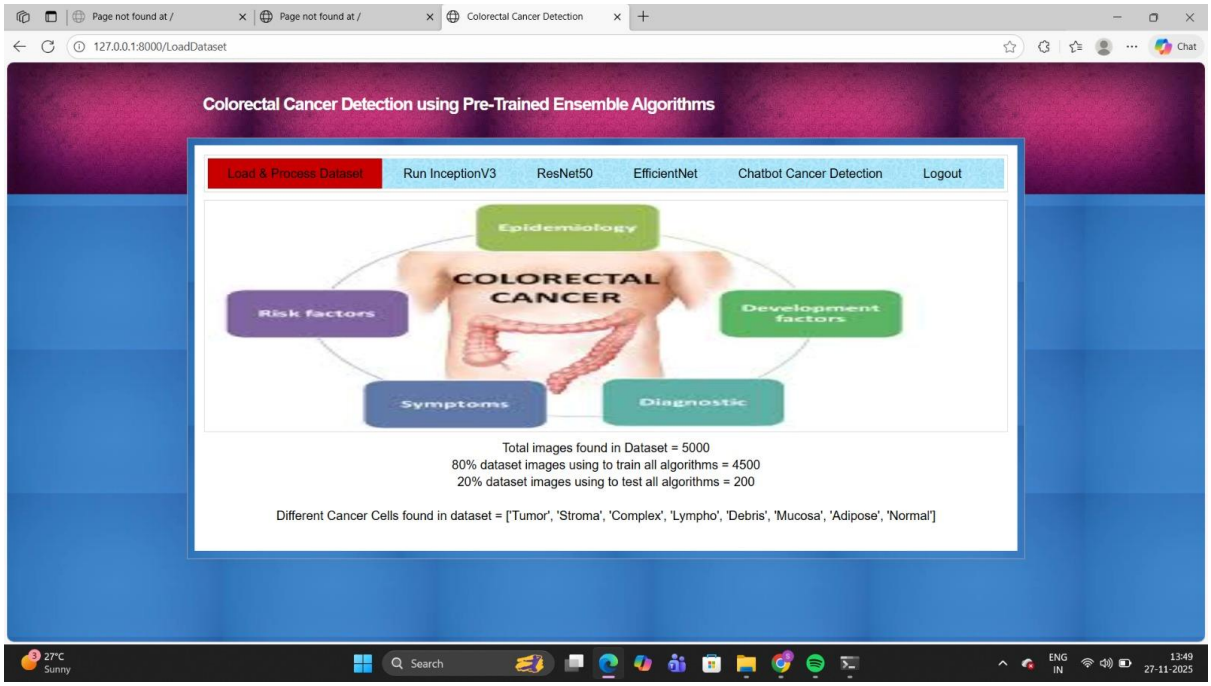


Figure 7.8 Information display page

This page displays the dataset summary including total images, training–testing split, and cell types present. It confirms that 5000 colorectal images were successfully loaded and processed.

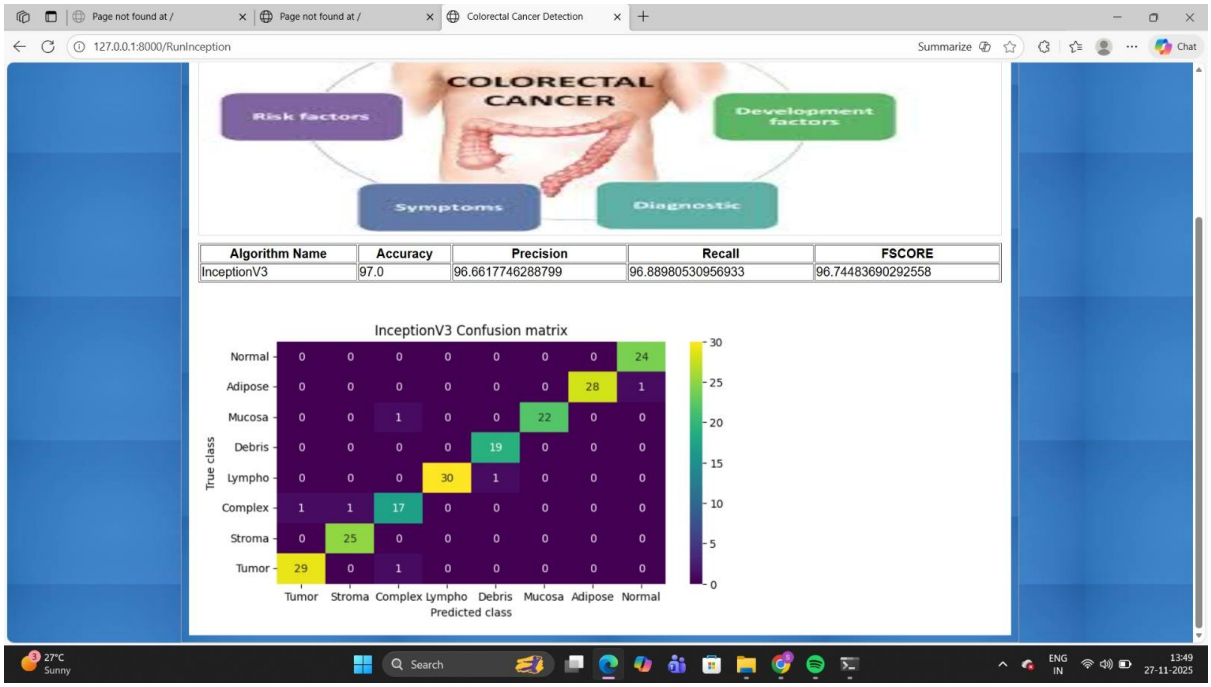


Figure 7.9 Run InceptionV3 page

This screen shows the InceptionV3 metrics along with its confusion matrix. The matrix demonstrates strong classification accuracy, with almost all samples correctly predicted.

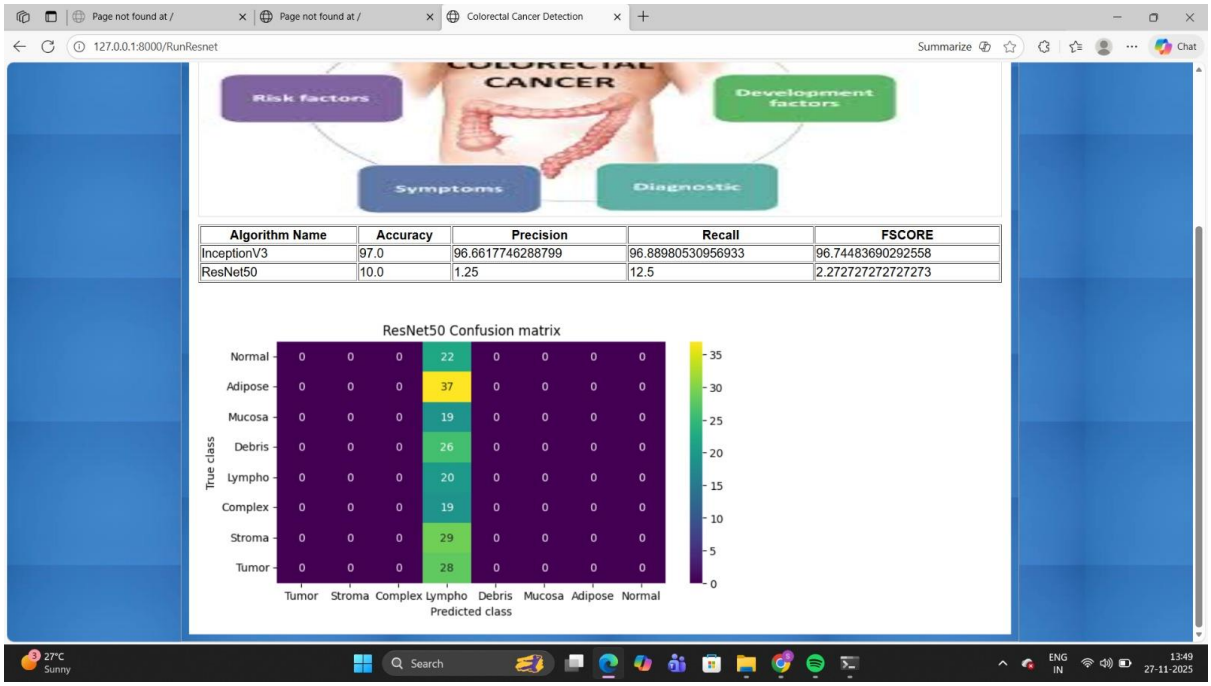
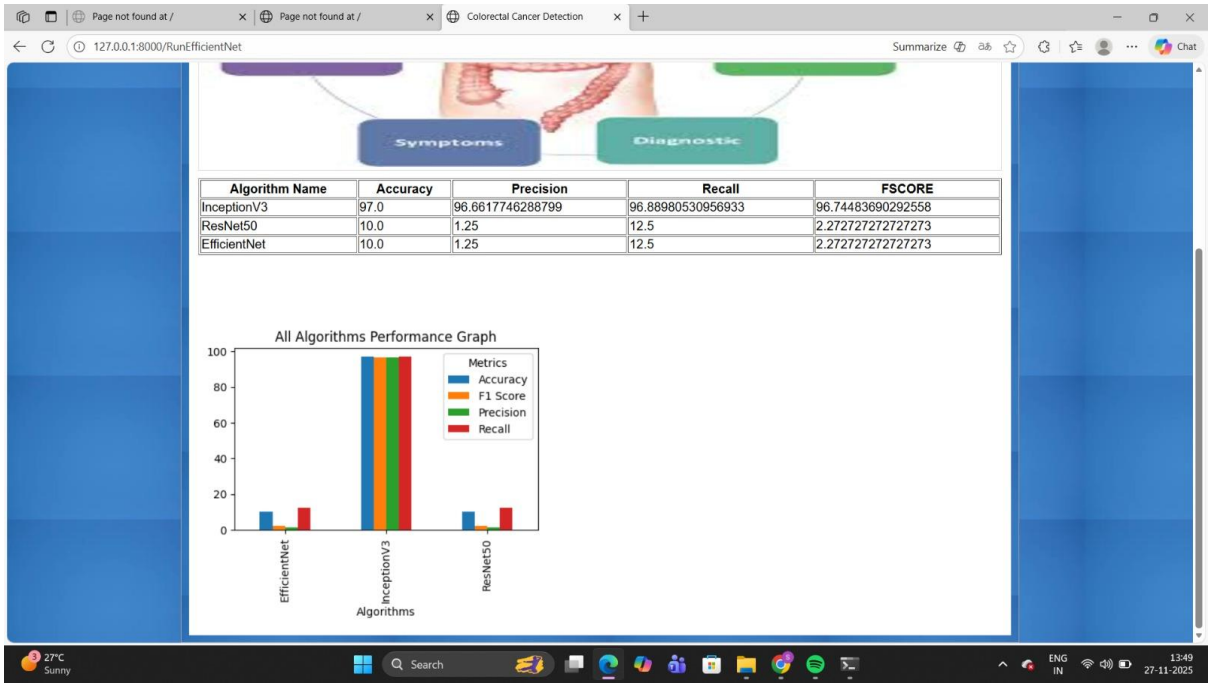


Figure 7.10 ResNet50 page

This image presents ResNet50's accuracy and metric table along with its confusion matrix. The matrix shows how the model misclassified most classes, indicating very low performance.



Algorithm Name	Accuracy	Precision	Recall	FSCORE
InceptionV3	97.0	96.6617746288799	96.88980530956933	96.74483690292558
ResNet50	10.0	1.25	12.5	2.272727272727273
EfficientNet	10.0	1.25	12.5	2.272727272727273

Figure 7.11 EfficientNet page

This screen shows the accuracy, precision, recall, and F1-score comparison of InceptionV3, ResNet50, and EfficientNet. A bar graph below visualizes how each algorithm performs across the four metrics.

8.CONCLUSION AND FUTURE ENHANCEMENT

8.1.CONCLUSION

Colorectal cancer detection remains a critical challenge in medical diagnostics, where accuracy and timeliness are vital. Traditional diagnostic methods, while reliable, are invasive and resource-intensive. Machine learning, particularly ensemble learning, presents a transformative opportunity to enhance early detection by combining multiple models to reduce error rates and improve predictive reliability. The proposed system, based on pre-trained ensemble algorithms (Random Forest, Gradient Boosting, XGBoost), aims to achieve superior performance compared to single-model approaches. By integrating structured data and possibly image-based datasets, the system can support clinicians with automated, explainable, and efficient diagnostic assistance. Ultimately, this approach has the potential to improve patient survival rates by enabling earlier intervention and more personalized treatment planning.

8.2.FUTURE ENHANCEMENT

In the future, this colorectal cancer detection system can be improved to make it even more helpful for doctors and patients. More medical data from different hospitals can be added so the system becomes more accurate for all types of people. The system can also use advanced deep learning models to better understand medical patterns and detect cancer even earlier. We can also connect this system directly in hospitals so that doctors get results immediately during checkups or colonoscopy tests. More patient information like medical history and genetic details can be included to give more correct predictions. The explanation of results can be improved so doctors clearly understand why the system predicted cancer. The system can be updated regularly with new data to keep it accurate for all future cases. Cloud support can also be added so that hospitals from different places can use the system easily. With these improvements, the system can help in faster detection of colorectal cancer and support better treatment and patient care in real life.

9.REFERENCES

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